Thereom 21

Suppose Ψ_0 is a set of atomic soft Interventions with Signature Σ. Let Ψ be the closure of Ψ_0 under function composition. Then $(\Psi, \circ)$ is an intervention algebra

- signature $\Sigma = (V, val)$ [variables & their value ranges]
- Ψ_0 is a hand picked collection of one variable mechanism replacements [menu of allowed swaps]
- subscript $0 \rightarrow$ starting set before we close it up.
- Ψ is the closure of Ψ_0 under function composition $\rightarrow$ Take Ψ_0 & throw every finite composite that can be formed

  Example: if $I, I' \in \Psi_0 \Rightarrow I \circ I'$ must $\in \Psi$. (& so on for longer chains)

*- Ψ is the smallest super set of Ψ_0 that you can't escape from by composing (algebra).

- * is needed because $(\Psi, \circ)$ is described as an Algebra & by definition, an algebra's operation must stay inside the set

  - Ψ_0 itself doesn't follow ↑, since composing 2 atoms targeting different variables gives a non-atomic element.
  - $(\Psi, \circ)$ is an intervention algebra $\rightarrow$ there exists some signature Σ^* such that $(\Psi, \circ)$ is isomorphic to $(\Phi, \circ)$, the hard interventions on Σ^* under composition.

    - In other words, this whole system of soft intervention composites is secretly just hard interventions on a clearly chosen variable space.

- From the proof:

  1. free algebra A^* is all finite sequences of atoms in $A = \Psi_0$. [sequence = to-do list of mechanism swaps]
  2. quotiented under $\approx$ to produce $(A^*/\approx, \odot)$

  apply equivalence relation $\approx$, which glues together sequences that differ only by (1) reordering swaps to different variables & (2) dropping a swap that a later swap overwrites.

- Quotient $A^*/\approx$ is the set of resulting equiv. classes (buckets) & $\odot$ is the induced operation.